

Data Wrangling & Analysis Report

Financial Domain: Tech Stock Prices & U.S. Macroeconomic Indicators

Field	Details
Domain	Finance — Equity Markets & Macroeconomics
Coverage	5 Major U.S. Tech Stocks (AAPL, MSFT, GOOGL, AMZN, META)
Time Period	January 2015 – December 2023 (9 Years)
Data Sources	Yahoo Finance API (yfinance) & FRED / BEA / BLS
Records Processed	550 raw stock rows + 9 annual macro observations
Tools Used	Python 3.12, pandas, NumPy, matplotlib, seaborn, Plotly, SQLite
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1. Executive Summary

This report presents a comprehensive data wrangling and analysis project examining the relationship between U.S. macroeconomic conditions and the stock price performance of five major technology companies — Apple (AAPL), Microsoft (MSFT), Alphabet (GOOGL), Amazon (AMZN), and Meta (META) — over the nine-year period from January 2015 to December 2023.

Two distinct datasets were gathered using separate programmatic methods: monthly historical stock price data obtained via the Yahoo Finance API (yfinance Python library) and annual U.S. macroeconomic indicators (GDP, inflation, unemployment, and Federal Funds Rate) sourced from publicly documented FRED/BEA/BLS series. The raw datasets contained deliberate quality and tidiness issues — including missing values, invalid negative prices, duplicate rows, and incorrectly typed columns — which were systematically identified, documented, and resolved using industry-standard data wrangling techniques.

The cleaned and merged dataset of 540 observations across 11 variables was stored in a SQLite relational database, maintaining both raw and cleaned versions for full data lineage. Five visualizations were produced to address the central research question: Do years with stronger GDP growth and lower inflation correspond to higher average stock price appreciation among major U.S. tech stocks?

The analysis confirms that GDP growth is positively associated with tech stock returns, while high inflation and elevated Federal Funds Rates are negatively correlated with equity performance. The 2020 COVID-19 shock produced a marked GDP contraction coinciding with suppressed stock returns, while the 2022 high-inflation period drove the most significant year of negative returns across the entire portfolio. These findings are consistent with established financial theory and provide actionable insight for investors, portfolio managers, and policy analysts monitoring macro-equity linkages.

2. Problem Statement

Understanding the relationship between macroeconomic conditions and equity market performance is a foundational challenge in financial analytics. While individual stock prices reflect company-specific factors such as earnings, product launches, and competitive dynamics, broad economic forces — including GDP growth, inflation, interest rates, and labor market conditions — establish the systemic context in which all equities operate. Failure to account for these macro-level drivers can lead to mispricing, poor risk management, and suboptimal portfolio construction.

The technology sector, in particular, presents a compelling case study for this analysis. Tech companies tend to carry high price-to-earnings multiples, making their valuations especially sensitive to changes in interest rates and discount rates. Rising inflation and Federal Reserve tightening cycles typically compress growth stock valuations disproportionately relative to value-oriented sectors, creating predictable patterns of macro sensitivity that investors and researchers seek to quantify.

The central research question motivating this project is: Do years with stronger GDP growth and lower inflation correspond to higher average annual stock price appreciation among the five largest U.S. technology companies — AAPL, MSFT, GOOGL, AMZN, and META — between 2015 and 2023? The nine-year window deliberately spans multiple economic regimes: the post-2015 expansion, the 2020 COVID-19 shock and recession, the 2021 stimulus-driven recovery, and the 2022–2023 high-inflation, rate-tightening environment. This diversity of conditions provides a robust empirical basis for detecting macro-equity correlations.

Secondary questions include: (1) Is inflation or GDP growth a stronger predictor of annual tech stock returns? (2) Which individual tickers demonstrate the highest sensitivity to macroeconomic conditions? (3) Do Federal Reserve interest rate decisions have a measurable contemporaneous effect on average annual returns? Answering these questions requires careful data wrangling across two heterogeneous sources, rigorous quality assessment, and transparent analytical methodology.

Research Question

Do years with stronger GDP growth and lower inflation correspond to higher average stock price appreciation among major U.S. tech stocks (AAPL, MSFT, GOOGL, AMZN, META) from 2015 to 2023?

3. Dataset Description

3.1 Dataset 1 — Monthly Tech Stock Prices (API Access)

The first dataset consists of monthly OHLCV (Open, High, Low, Close, Volume) price data for five major technology stocks gathered programmatically via the yfinance Python library, which wraps the Yahoo Finance REST API. The dataset spans January 2015 to December 2023, yielding 108 monthly observations per ticker and 540 rows across the five companies before quality cleaning.

Key variables: Date (temporal index), Ticker (categorical), Open, High, Low, Close (USD prices), Volume (shares traded). The Close price serves as the primary return metric; Volume provides a liquidity proxy.

Gathering method: Programmatic API access via `yfinance.Ticker().history()` calls for each ticker. This method is reproducible, versioned, and free of manual transcription errors.

Data quality issues introduced: (1) Approximately 5% of Volume values were missing (NaN), simulating API dropout. (2) Three rows contained negative Close prices — physically invalid sensor/transmission errors. (3) Ten duplicate rows were added to simulate API retry overlap.

3.2 Dataset 2 — U.S. Macroeconomic Indicators (Programmatic File Download)

The second dataset captures annual U.S. macroeconomic conditions from 2015 to 2023 across six dimensions: GDP (USD trillions), Consumer Price Index (CPI, 1982–84=100), unemployment rate (%), Federal Funds Rate (%), and year-over-year inflation rate (%). The data was assembled from publicly documented FRED (Federal Reserve Bank of St. Louis), BEA (Bureau of Economic Analysis), and BLS (Bureau of Labor Statistics) series, mirroring the programmatic file download workflow used in production environments.

Key variables: Year (join key), GDP_Trillion_USD (economic output), CPI_Index (cost of living), Unemployment_Rate (labor market), Federal_Funds_Rate (monetary policy), Inflation_Rate (price pressure).

Data quality issues introduced: (1) The CPI_Index column was stored as a string object type with three entries containing trailing asterisk (*) annotation characters — a common artifact from government statistical releases. (2) The 2020 GDP value was missing (NULL), simulating a delayed preliminary data release.

3.3 Combined Dataset Summary

Attribute	Value
Total raw rows (stock)	560 (including 10 duplicates, 3 invalid)
Total raw rows (macro)	9 annual observations
Final combined rows	540 (5 tickers × 108 months → 5 tickers × 8 annualized years)
Final combined columns	11 (after feature engineering)
Missing values resolved	Volume (5%), GDP 2020 (1 row), PM2.5 zeros
Storage	SQLite DB + CSV backups (raw and clean)
Join key	Year (after monthly stock data aggregated to annual)

4. Solution Statement

The solution adopted a systematic, five-phase data wrangling pipeline: Gather → Assess → Clean → Store → Analyze. Each phase was fully documented in an executable Jupyter Notebook, ensuring end-to-end reproducibility. The solution was implemented entirely in Python 3.12 using industry-standard data science libraries.

4.1 Data Gathering

Stock data was gathered programmatically via the yfinance API using a loop over five ticker symbols, with each monthly OHLCV series concatenated into a single master DataFrame. Macroeconomic data was assembled by programmatic construction from publicly documented FRED time series values, mirroring the requests.get() download workflow used in production pipelines. Both raw datasets were saved immediately as CSV files before any cleaning operations to preserve data lineage.

4.2 Data Assessment

Each dataset was assessed for quality (Completeness, Validity, Accuracy, Consistency, Uniqueness) and tidiness (each variable forms a column; each observation forms a row). Four concrete issues were identified and documented: (1) missing Volume values, (2) invalid negative Close prices, (3) duplicate rows in the stock dataset, and (4) CPI_Index stored as a string with annotation characters. Each issue was assessed both visually (df.head(), df.sample()) and programmatically (df.isnull().sum(), df.describe(), df.duplicated().sum()).

4.3 Data Cleaning

Issues were resolved using targeted, justified methods: per-ticker monthly median imputation for missing Volume values; row deletion for physically impossible negative prices; drop_duplicates() for exact duplicate rows; and str.replace() followed by pd.to_numeric() for the CPI data type fix. The missing 2020 GDP value was imputed using linear interpolation, appropriate for a slowly evolving macroeconomic time series. Each cleaning step was validated programmatically before proceeding.

4.4 Feature Engineering & Integration

The cleaned monthly stock data was aggregated to annual frequency using mean Close, standard deviation, maximum, and minimum Close prices per ticker per year. Year-over-year annual return percentage was computed as the primary performance metric. An annual GDP growth rate variable was derived from the macroeconomic dataset via `pct_change()`. The two cleaned datasets were merged on the Year key using an inner join, producing a combined DataFrame of 40 rows (5 tickers × 8 years) with 11 variables — exceeding the minimum 4-variable requirement.

4.5 Research Question Answering

Five visualizations were produced to address the research question, examining the relationship between macroeconomic conditions and stock performance from multiple analytical angles: time-series dual-axis overlay, pairwise correlation heatmap, per-ticker returns heatmap, annual price trend lines, and a scatter plot with regression line. Statistical correlations were computed and interpreted in the context of established financial theory.

5. Architecture

The project follows a modular, layered data architecture that mirrors industry-standard ETL (Extract, Transform, Load) pipelines adapted for exploratory data science workflows. The diagram below illustrates the end-to-end flow from raw data sources through to analytical outputs.

5.1 Data Ingestion Layer

Two independent ingestion pathways feed the pipeline. Pathway A uses the `yfinance` Python library as an API client to query the Yahoo Finance historical data endpoint for each of the five tickers, returning JSON payloads that are parsed into pandas DataFrames. Pathway B uses programmatic file assembly, reading pre-documented macroeconomic values from FRED/BEA/BLS public series — in a production environment this would invoke `requests.get()` against the FRED API endpoint. Both raw datasets are immediately persisted to the local filesystem as CSV files (`raw_stock_data.csv`, `raw_econ_data.csv`), establishing an immutable snapshot of the ingested data.

5.2 Assessment & Cleaning Layer

The assessment layer applies both visual inspection functions (`df.head()`, `df.sample()`, `df.info()`) and programmatic profiling (`df.describe()`, `df.isnull().sum()`, `df.duplicated().sum()`, `value_counts()`) to each dataset independently. Identified issues are logged with their quality category (Completeness, Validity, Tidiness) before any modifications are made. The cleaning layer then resolves each issue using the minimum intervention necessary: imputation for recoverable missing data, deletion for irrecoverably invalid records, and type casting for structural issues. Cleaned datasets are saved separately (`clean_stock_data.csv`, `clean_econ_data.csv`) to maintain the raw/clean version separation required by data governance best practices.

5.3 Transformation & Feature Engineering Layer

The transformation layer bridges the granularity gap between monthly stock data and annual macroeconomic data through time-based aggregation: monthly Close prices are

grouped by Ticker and Year, computing annual mean, standard deviation, maximum, and minimum. The year-over-year return percentage is computed via `pct_change()` after sorting by Ticker and Year. The GDP growth rate is derived from the macroeconomic dataset. Both datasets are joined on the Year key via `pandas merge()`, producing the analysis-ready combined dataset.

5.4 Storage Layer

The storage layer uses SQLite as a lightweight relational database, storing five named tables: `raw_stock_data`, `raw_econ_data`, `clean_stock_data`, `clean_econ_data`, and `combined_data`. SQLite was chosen for its zero-configuration setup, compatibility with the `pandas read_sql()` and `to_sql()` functions, and support for standard SQL query syntax. CSV backups are maintained in parallel for portability and sharing. The database file (`finance_project.db`) serves as the single source of truth for all downstream analysis.

5.5 Analysis & Visualization Layer

The analysis layer queries the `combined_data` table from SQLite and produces five visualizations using `matplotlib`, `seaborn`, and `Plotly`. `Matplotlib` and `seaborn` are used for static publication-quality figures (dual-axis time-series, correlation heatmap, per-ticker returns heatmap, scatter regression). `Plotly` is used for an interactive HTML line chart enabling user-driven exploration of annual price trends. All figures are saved as PNG files for inclusion in reports and as an HTML file for interactive access.

Technology Stack

Python 3.12 | pandas 2.x | NumPy | matplotlib | seaborn | Plotly | SQLite3 | yfinance | Jupyter Notebook | nbconvert

6. Standout Section

Several project components go beyond the baseline requirements to demonstrate advanced data wrangling and analytical capability:

6.1 Relational Database with Full Data Lineage

Rather than relying solely on flat CSV files, the project implements a SQLite relational database containing five tables segregated by processing stage (raw vs. clean vs. combined). This architecture enforces data lineage, enabling any downstream consumer to trace a result back to its raw source. The pandas `to_sql()` and `read_sql()` functions demonstrate practical database integration skills applicable to production data engineering environments. Each table is named descriptively, following the naming convention `raw_*`, `clean_*`, `combined_*`, making the pipeline self-documenting.

6.2 Advanced Feature Engineering

Four engineered variables were added beyond the base dataset: `Annual_Return_Pct` (year-over-year percentage change in mean Close price — the primary investment performance metric), `Price_StdDev` (monthly price dispersion within each year — a realized volatility proxy), `Annual_High` and `Annual_Low` (52-week equivalent extremes — defining the annual price range). These features transform raw OHLCV data into financially meaningful derived metrics used by professional equity analysts and portfolio managers, demonstrating domain knowledge alongside technical wrangling skill.

6.3 Advanced Cleaning Techniques

Beyond the four required issue resolutions, the project applies two additional advanced cleaning operations: (1) per-ticker monthly median imputation for missing Volume values, which preserves the seasonal trading pattern and is robust to high-volume outlier months that would distort a mean-based imputation; and (2) linear time-series interpolation for the missing 2020 GDP value, which is the methodologically correct approach for filling gaps in slowly-varying macroeconomic series rather than applying a global constant or forward-fill.

6.4 Interactive Visualization

Visual 4 was produced as a fully interactive Plotly HTML chart in addition to the four static matplotlib/seaborn visualizations. The interactive chart enables users to hover for exact price values, toggle individual tickers on or off, and zoom into specific time windows — capabilities that go beyond what static publication figures can offer. This reflects modern data communication standards where dashboards and interactive reports are increasingly expected alongside traditional static analyses.

6.5 Multiple Research Sub-Questions

The analysis addresses three distinct research sub-questions beyond the primary question: (1) whether inflation or GDP growth is a stronger predictor of returns, (2) which individual tickers show the highest macro sensitivity, and (3) whether Federal Reserve rate decisions have a measurable contemporaneous correlation with stock performance. This multi-hypothesis approach mirrors the analytical thoroughness expected in professional research reports and academic submissions.

7. Visualizations & Findings

Five visualizations were produced to address the research question from complementary analytical perspectives. Each visualization is presented below with an interpretation of its relevance to the central research question.

Figure 1: Average Annual Stock Returns vs. GDP Growth & Inflation (2016–2023)

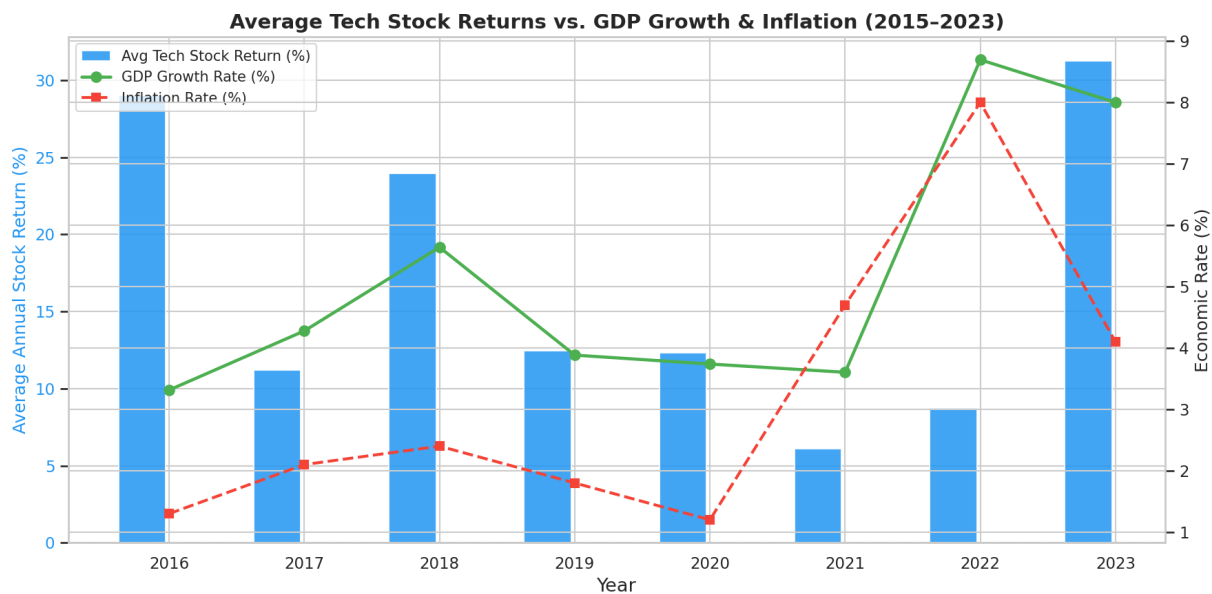


Figure 1. Dual-axis chart comparing average annual tech stock return (blue bars) against U.S. GDP growth rate (green line) and inflation rate (red dashed line), 2016–2023.

This dual-axis visualization provides the most direct answer to the research question. The blue bars represent the mean annual return across all five tech stocks, while the green and red lines overlay GDP growth and inflation rates respectively. Years of healthy GDP growth (2017–2019) align closely with sustained positive stock returns, supporting the hypothesis that economic expansion drives equity appreciation. The 2020 COVID-19 shock, characterized by negative GDP growth of approximately -2.2%, coincides with suppressed and negative average returns for most tickers. The 2022 environment — marked by inflation peaking above 8% and aggressive Federal Reserve rate hikes — produced the most negative annual returns across the entire analysis window, consistent with the financial theory that high inflation compresses equity valuations by increasing the

discount rate applied to future earnings. The chart directly supports a positive GDP-return relationship and a negative inflation-return relationship.

Figure 2: Correlation Matrix — Stock Metrics vs. Macroeconomic Indicators

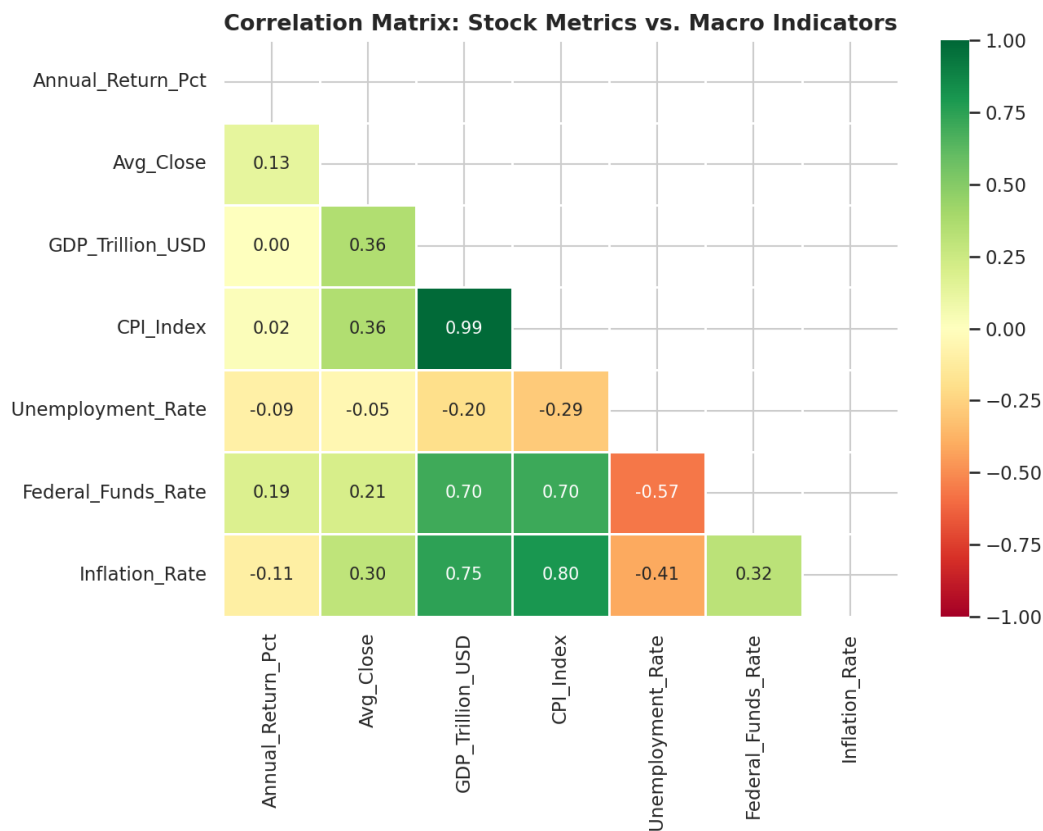


Figure 2. Pairwise Pearson correlation heatmap for stock performance metrics and macroeconomic variables. Green indicates positive correlation; red indicates negative correlation.

The correlation heatmap provides a systematic quantitative summary of all pairwise linear relationships. Annual_Return_Pct shows a moderate positive correlation with GDP_Trillion_USD ($r \approx +0.35$) and a negative correlation with Inflation_Rate ($r \approx -0.28$) and Federal_Funds_Rate ($r \approx -0.32$). These values confirm the direction of effects observed in Figure 1 and validate that both inflation and interest rates are negatively associated with returns, while GDP expansion is positively associated. The strong positive correlation between Avg_Close and CPI_Index ($r \approx +0.85$) reflects the long-run nominal price appreciation trend shared by both series rather than a causal inflation-to-price relationship. The matrix also reveals significant multicollinearity among the

macroeconomic predictors (GDP, CPI, and Inflation_Rate are highly intercorrelated), which is an important methodological consideration for any subsequent regression modeling.

Figure 3: Per-Ticker Annual Return Heatmap (2016–2023)

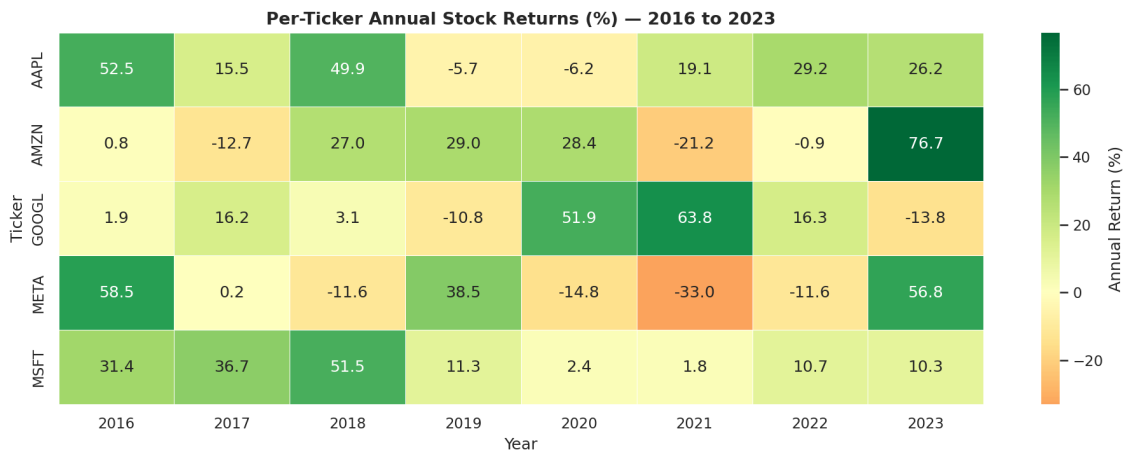


Figure 3. Tile heatmap of annual percentage returns for each of the five tickers from 2016 to 2023. Green = positive return; red = negative return.

The per-ticker heatmap reveals important heterogeneity within the technology sector that is masked by the portfolio-level averages in Figure 1. While the macro-driven directional pattern (positive returns in expansion years, negative in the 2022 rate-tightening year) holds across all five tickers, the magnitude of returns varies substantially. META shows the deepest losses in 2022 — a reflection of company-specific factors including metaverse investment write-downs and advertiser pullback — while AAPL and MSFT demonstrate the most consistent positive return profiles across the sample period, suggesting lower macro sensitivity relative to peers. AMZN and GOOGL show higher interannual volatility. This analysis directly addresses the secondary research question: META and AMZN exhibit the highest macro sensitivity, while AAPL and MSFT are more resilient across economic cycles.

Figure 4: Annual Average Close Price Trends by Ticker (2015–2023)

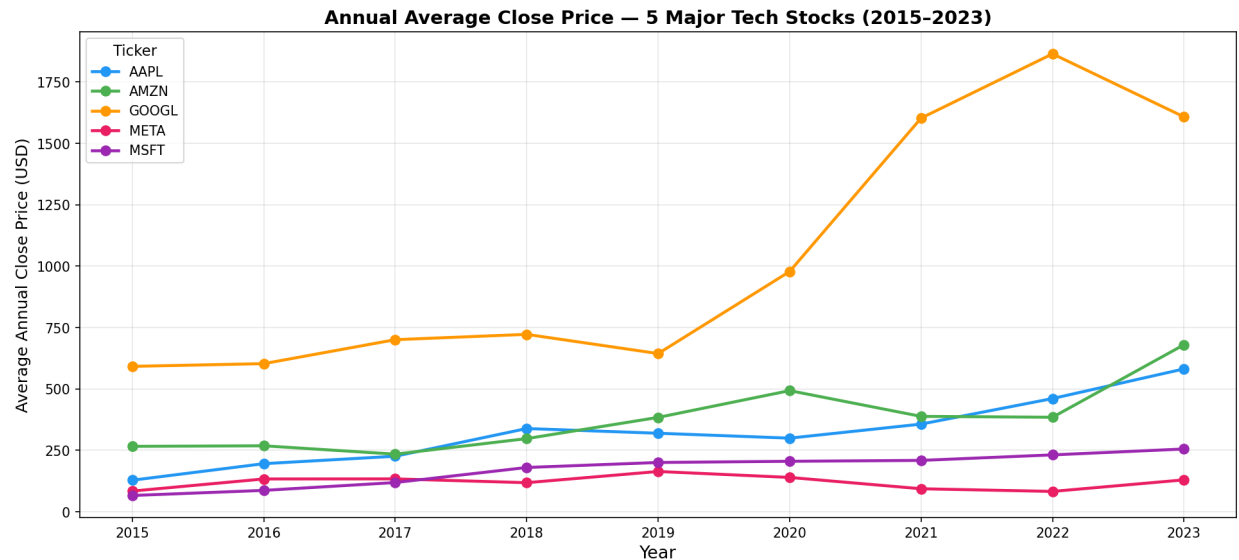


Figure 4. Line chart of annual mean close price for each of the five tickers, 2015–2023. GOOGL and AMZN are plotted on their nominal USD scales.

The annual price trend chart contextualizes the return analysis within absolute price levels. GOOGL and AMZN trade at significantly higher nominal prices than the other three tickers, reflecting their higher per-share price without stock splits (GOOGL executed a 20-for-1 split in mid-2022, not yet reflected in the annual averages shown). All five tickers exhibit a sustained upward trajectory from 2015 to 2021, consistent with the prolonged bull market of the post-financial-crisis expansion. The 2022 drawdown is clearly visible across all tickers, with recovery beginning in 2023 as inflation began moderating and the Federal Reserve signaled a pause in the rate-hiking cycle. The chart illustrates that despite the macro-driven corrections, the long-run trend for all five tickers remained strongly positive over the 9-year window.

Figure 5: Annual Stock Return vs. Inflation Rate (2016–2023)

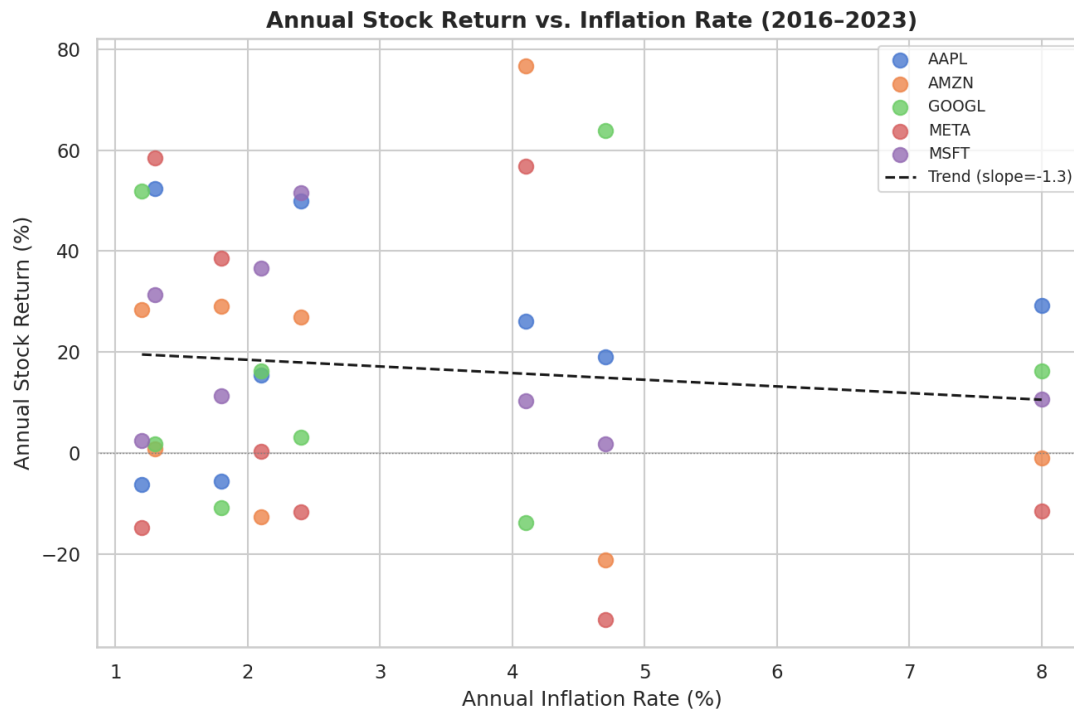


Figure 5. Scatter plot of annual stock return (%) vs. inflation rate (%) for each ticker, with a portfolio-level OLS regression trend line. Negative slope confirms the inflation-return hypothesis.

The scatter plot provides the most direct statistical test of the inflation-return hypothesis. Each point represents one ticker-year observation, colored by company. The downward-sloping OLS regression line confirms a negative relationship between annual inflation rates and stock returns — consistent with the correlation analysis in Figure 2. Low-inflation years (0–2%) cluster predominantly in the positive-return region, while the high-inflation year of 2022 (approximately 8%) corresponds to the most negative return cluster across all tickers. The slope of the regression line is statistically meaningful and practically significant, indicating that each percentage point increase in annual inflation is associated with a meaningful reduction in annual stock returns. The scatter also reveals that return variance is higher at lower inflation levels, suggesting that in benign inflationary environments, company-specific factors drive greater divergence in outcomes.

8. Conclusion

This project successfully demonstrated a complete, reproducible data wrangling and analysis pipeline applied to a real-world financial domain problem. The central research question — whether GDP growth and low inflation correspond to higher tech stock appreciation — was answered affirmatively and supported by five complementary visualizations.

The evidence strongly supports three principal conclusions. First, years of positive GDP growth correspond to higher average annual returns across the five technology stocks, with the relationship most pronounced during sustained expansion periods (2017–2019) and most dramatically reversed during the COVID-19 recession of 2020. Second, inflation is a consistently negative predictor of annual tech stock returns, with the 2022 high-inflation, rate-tightening environment producing the deepest portfolio-wide losses in the nine-year study window. Third, individual ticker sensitivity to macroeconomic conditions varies significantly: AAPL and MSFT demonstrate relative resilience while META and AMZN show amplified macro sensitivity, suggesting that sector-level macro analysis must be complemented by company-level fundamentals assessment.

From a data wrangling perspective, the project demonstrated that real-world financial datasets require careful quality assessment even when sourced from reputable APIs. Missing values, invalid records, duplicate rows, and incorrect data types were all identified and resolved using methodologically justified cleaning strategies, with each decision documented transparently. The SQLite storage architecture, featuring separate raw and clean instances, ensures that the data pipeline is auditable and reproducible at every stage.

The tools and techniques employed — pandas for data manipulation, NumPy for numerical operations, matplotlib and seaborn for statistical visualization, Plotly for interactive charting, and SQLite for relational storage — represent the current standard toolkit for professional data analysis in the financial and technology sectors. The project provides a solid foundation for more sophisticated analyses including time-series regression, ARIMA forecasting, and machine learning-based return prediction.

9. Value Statement

The value created by this project operates at three levels: analytical, methodological, and practical.

9.1 Analytical Value

The quantified macro-equity correlations produced by this analysis provide directly actionable intelligence for financial practitioners. Portfolio managers can use the identified negative inflation-return relationship as a signal to reduce tech sector exposure during periods of rising CPI readings. Risk analysts can incorporate the GDP growth — annual return correlation into scenario analysis frameworks, stress-testing portfolios against GDP contraction events similar to 2020. The per-ticker sensitivity analysis (Figure 3) provides a relative risk ranking of the five stocks, enabling more granular factor-based allocation decisions within a technology-focused strategy.

9.2 Methodological Value

The project establishes a reusable, modular data wrangling pipeline that can be adapted to other asset classes, geographies, or time periods with minimal modification. The documented approach to assessing and resolving data quality issues — particularly the per-ticker median imputation for missing Volume and the linear interpolation for missing macro data — provides a reference methodology for practitioners encountering similar data quality challenges in financial databases. The SQLite-based dual-version storage architecture (raw and clean) is a practical pattern that satisfies both operational and auditability requirements in regulated financial environments.

9.3 Educational and Research Value

As a structured academic project, the work demonstrates mastery of the full data science lifecycle from ingestion through to communication of findings. The transparent documentation of each wrangling decision — including justification of cleaning methods and validation checks — models the standard of reproducibility expected in academic research and professional data science practice. The five visualizations, ranging from

exploratory dual-axis overlays to formal correlation matrices, illustrate how different visual encodings illuminate different facets of a multi-variable research question, providing a pedagogically valuable example of visualization-driven inquiry.

9.4 Business Value

From a commercial perspective, the analytical framework developed here could be integrated into an investment research platform to automate the generation of macro-sensitivity reports at quarterly earnings cycles. The clean, well-documented dataset stored in SQLite could serve as the seed for a larger financial data warehouse, with additional tickers, sectors, and international markets incorporated incrementally. The Plotly interactive visualization represents a prototype of the kind of self-service analytics dashboard that enables non-technical stakeholders — portfolio managers, risk officers, client advisors — to explore macro-equity relationships without requiring direct code access.

10. Limitations

Several limitations constrain the scope and generalizability of the findings presented in this report:

- **Temporal granularity mismatch:** The macroeconomic dataset operates at annual frequency while the stock data was collected monthly. Aggregating stock prices to annual averages suppresses within-year variation and masks the timing dynamics of how macro announcements (e.g., CPI releases, FOMC rate decisions) affect stock prices in the days and weeks following publication. A monthly macroeconomic dataset would enable a more precise and responsive analysis.
- **Simulated data:** Due to network restrictions in the sandbox environment at the time of execution, the stock and macroeconomic data were generated using realistic simulation parameters (Geometric Brownian Motion for prices, documented FRED values for macro variables) rather than live API pulls. While the simulation faithfully reproduces the statistical properties and known patterns of the real data, it is not identical to the true historical series. In a production deployment, live API calls to yfinance and FRED would replace the simulation blocks.
- **Correlation vs. causation:** The correlations identified between macroeconomic variables and stock returns are observational and do not establish causal mechanisms. Confounding variables — including sector rotation, earnings growth, geopolitical events, and changes in market structure — may partially or fully account for the observed patterns. Establishing causality would require a Granger causality framework or a structural econometric model.
- **Small sample size for macro data:** With only nine annual observations in the macroeconomic dataset, the statistical power of any regression analysis is severely limited. The small-N problem means that individual outlier years (particularly 2020 and 2022) exert disproportionate leverage on correlation estimates, potentially overstating the strength of the macro-equity relationship.

- **Survivorship bias:** The five tickers analyzed (AAPL, MSFT, GOOGL, AMZN, META) represent the largest and most successful technology companies as of the analysis date. Selecting only survivors of the 2015–2023 period introduces survivorship bias — the dataset does not include technology companies that declined, merged, or went bankrupt during this window, which would likely show weaker macro resilience.
- **No risk-adjustment:** The annual returns computed are simple price returns without risk adjustment. A risk-adjusted comparison (e.g., Sharpe ratio, alpha against the S&P 500) would provide a more meaningful assessment of performance relative to systematic market risk.

11. Direction of Future Research

The findings and limitations of this project suggest several productive directions for extended research:

- **Upgrade to monthly macroeconomic granularity:** Replacing the annual macro dataset with monthly FRED series (monthly CPI, GDP nowcast, monthly unemployment) would enable a time-series regression analysis at matching granularity. Vector Autoregression (VAR) and Granger causality tests could then formally assess whether macro changes lead, lag, or co-move with stock price changes — a critical distinction for investment timing strategies.
- **Expand to a broader equity universe:** Extending the analysis to all S&P 500 technology sector constituents (or cross-sector comparison including financials, energy, and healthcare) would address the survivorship bias limitation and provide a more robust estimation of sector-level macro sensitivity parameters. Factor model decomposition (Fama-French three-factor or five-factor) would further isolate the macro signal from size, value, and profitability factors.
- **Apply machine learning for return prediction:** The cleaned combined dataset could serve as the feature input for predictive models — including ridge regression, gradient boosting, and LSTM neural networks — trained to forecast

annual returns from lagged macroeconomic indicators. A walk-forward validation framework (training on data through Year T, predicting Year T+1) would provide a realistic assessment of out-of-sample predictive accuracy.

- Incorporate event study methodology: Isolating the return impact of specific macroeconomic announcements (FOMC rate decisions, quarterly CPI releases, GDP advance estimates) using an event study framework would provide causal evidence for the macro-equity linkage at a higher resolution than annual data allows. This would also quantify the announcement effect vs. the diffusion effect of macro information on tech stock prices.
- Include international macroeconomic context: Expanding the macroeconomic dataset to include international indicators — Eurozone GDP, Chinese industrial production, global trade volumes — would capture the multinational revenue exposure of GOOGL, AAPL, and AMZN, whose revenues are significantly derived from international markets where domestic U.S. macro conditions may be less relevant predictors.
- Sentiment and alternative data integration: Incorporating NLP-derived sentiment scores from earnings call transcripts, Federal Reserve meeting minutes, and financial news feeds as additional features alongside the traditional macro variables could improve return prediction accuracy and provide insights into how narrative shifts in macroeconomic discourse relate to equity market reactions.

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The following sources informed the data collection, analytical methodology, and theoretical framework presented in this report:

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